higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

INTRODUCTION TO SYSTEMS DEVELOPMENT
(First Paper)
NQF LEVEL 2

12 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1 C
1.2 B
1.3 B
1.4 A
1.5 C
1.6 B
1.7 A
1.8 B
1.9 A
1.10 C

(10 × 1) **[10]**

QUESTION 2

2.1 False
2.2 True
2.3 False
2.4 False
2.5 True
2.6 True
2.7 True
2.8 True
2.9 False
2.10 True

(10 × 1) **[10]**

QUESTION 3

3.1
- Usability
- Functionality
- Productivity
- Compatibility (4)

3.2
- Compliers
- Interpreters
- Assemblers (Any 2 × 1) (2)

3.3 Backward compatibility is downward compatibility which means to be able to interact with older versions of the same product, while forward compatibility means to be able to accept instructions or data in later versions. (2 + 2) (4)

3.4
- Multi-user
- Multiprocessing
- Multitasking
- Multithreading
- Real-time (5)

3.5 A device driver acts as a communication device between computer programs and hardware or peripheral devices. (1)
[16]

QUESTION 4

4.1 **Advantages**
- User-friendly
- Easy to maintain code
- Create complex programs
- Structured programming
- Portable and generally platform independent
- Faster development time
- Better control over data and software (Any FOUR)

Disadvantages
- Consume a lot of memory
- Additional process of compilation needs more CPU time
- Need to recompile every time the source code changes
- Sacrifice efficient code for more easily written code
- Debugging more difficult (Any FOUR)

(4 + 4) (8)

4.2 4.2.1 A class is the blueprint from which objects are built.✓ Classes are pieces of code that describe the state and behaviour of an object.✓ A class is an ADT (abstract data type).✓ (3)

4.2.2
- The data stored within an object represents the state of the object and is called attributes.
- Methods represent the behaviour of an object and are contained in functions and subprocedures.

OR

- Attributes describe what an object will look like, while methods are actions an object can perform. (2)

4.2.3
- Weight
- Height (2)

4.2.4
- CalcBMI()
- DisplayStatus() (2)

4.3 4.3.1
- Safe and easy to use
- Faster, easier, more systematic programming
- Robust and already proven
- Well-documented
- Thoroughly tested (5)

4.3.2
- C#
- C++
- Smalltalk
- Delphi
- Java (Any 3 × 1) (3)

[25]

QUESTION 5

5.1
- Software-requirement analysis
- System analysis and design
- Code generation
- Software testing
- Maintenance

OR

- Understand the problem.
- Devise possible solutions.
- Analyse the solutions and choose the best solution.
- Test the solution and apply the changes.
- Test and implement the solution. (5)

5.2 To represent characters as numbers (1)

5.3
5 – 9 \ 8 + 17 mod 3 * 2✓
5 – 9 \ 8 + 17 mod 6✓
5 – 1 + 17 mod 6✓
5 – 1 + 5✓
4 + 5 = 9✓ (5)

[11]

QUESTION 6

6.1
- Syntax errors
- Warning error messages
- Logical errors
- Run-time errors (4)

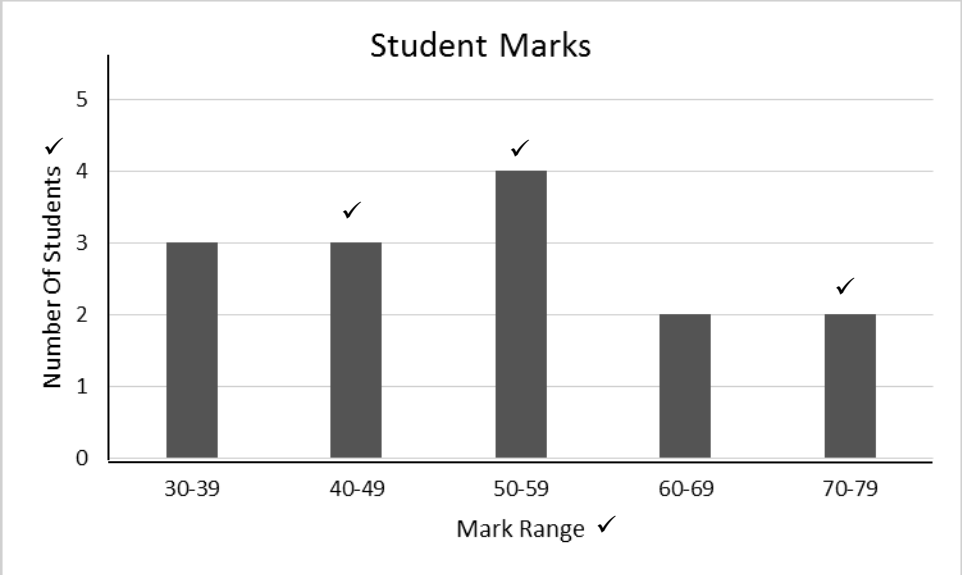
6.2
- TextBox
- Label
- CommandButton
- RadioButton
- CheckBox
- GroupBox
- ListBox
- ComboBox (Any 4 × 1) (4)

6.3 6.3.1

Class (Mark range)	Number of students
30-39	3
40-49	3
50-59	4
60-69	2
70-79	2

(5)

6.3.2



(5)
[18]

QUESTION 7

7.1
- Component testing is conducted to verify the implementation of the design for one software element or a collection of software elements.
- Integration testing is an orderly progression of testing in which various software and/or hardware elements are integrated and tested.
- System testing is the process of testing an integrated hardware and software system to verify that the system meets its specified requirements.
- Acceptance testing is formal testing conducted to decide whether or not the system satisfies its acceptance criteria and to enable the customer to determine whether or not to accept the system. (Any 2 × 2) (4)

7.2 Feasibility is an evaluation to determine whether a project or product will be a profitable investment. Viability is the ability of the project or product to perform the tasks required and to adapt to the changed environment. (2 + 2) (4)

7.3
- Detects and corrects errors while still inexpensive to correct
- Improves the quality of the product
- Provides a good resource for improving software products
- Establishes a good metrics database of planning as well as reliability and failure predictions (Any 2 × 1) (2)

[10]

TOTAL: 100

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